

BUTACU IONEL-CATALIN

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SUMMARY

C++ Software Engineer with 3 years of production experience delivering features end-to-end on a large-scale industrial CAD/CAM platform. Background in control systems and applied informatics. Currently completing an M.Sc. in Artificial Intelligence, with applied work in machine learning and embedded systems.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems

Software Engineer

Iași, RO

June 2023 – Present

- Designed and implemented features across Qt UI, networking, model/view, and event-handling layers of a production CAD/CAM platform.
- Translated complex product requirements into scoped, deliverable tasks, consistently meeting sprint commitments.
- Reduced technical debt through targeted refactoring, improving build stability and long-term codebase maintainability.
- Diagnosed and resolved production crashes and regressions through systematic root cause analysis.
- Participated in the full Scrum lifecycle: sprint planning, estimation, code reviews, sprint demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Developed UI components from design specifications and integrated them into existing production views.
- Delivered pixel-accurate UI and extended feature configurability via import/export workflows.
- Applied full Git workflow: branching strategies, commit conventions, merge requests, code reviews, and CI/CD pipelines.

Special Telecommunications Service (STS)

IT Support Staff — Electoral IT Systems

Botoșani, RO

August 2020 – September 2020

- Provided technical support for electoral IT systems (e.g., SIMPV/SICPV).
- Assisted polling-station staff with application usage and hardware operation.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25)

CORE SPECIALTIES

Languages: C++17/20, C#, Python, SQL

Frameworks: Qt 5/6, .NET 8, STL

Engineering: Event-driven Architecture, Design Patterns, Multithreading, API Integration, Performance Profiling, CI/CD

Tooling: MSVC, CMake, Visual Studio, GitLab, Docker, NuGet, SonarQube, Jira

EDUCATION

Gheorghe Asachi Technical University of Iași

Master of Science in Artificial Intelligence

Iași, RO

October 2024 - June 2026

Advanced Coursework: Neural Networks, Machine Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Automatic Control and Applied Informatics

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Database Management Systems, Real-Time Programming, Systems Theory, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Neural Networks & Fuzzy Logic, Knowledge-Based Systems

PROJECTS

M.Sc. Thesis: SSM for Motion Generation	2026
Developing S4 and Mamba-based architectures for temporally consistent text-to-motion generation (KIT-ML dataset). Incorporated physics constraints to eliminate motion jitter and improve long-range dependency modelling.	
Intelligent IDS: SQL Injection Detection	2026
Developed a clustering-based rule generator for SQL injection detection, embedded within a multi-component IDS alongside an adversarial attack agent and a classification model. Each module evaluated under two independent benchmarking strategies.	
HandSense: NPU Gesture Recognition	2025
End-to-end YOLOv5n deployment for real-time gesture recognition (HaGRID dataset). Optimised for NPU/CPU with TFLite INT8 quantisation, CLAHE preprocessing, and temporal smoothing.	
PharmaRAG: Local LLM Assistant	2025
Offline RAG system with FAISS vector search and Mistral-7B for medication retrieval. Chain-of-thought prompting for drug interaction analysis and contraindication detection.	
HeuristicOpt: Optimization Framework	2025
Python framework with Genetic, Evolutionary, and Stochastic solvers for constraint satisfaction. Benchmarked across convergence speed and solution quality in large-scale search spaces.	
WiDS: fMRI Brain-Age Prediction (8th National)	2025
ML pipeline for fMRI-based brain-age prediction on neuroimaging connectivity data (~20k features). Applied ElasticNet with spectral feature selection and gender-stratified training.	

VOLUNTEER EXPERIENCE

LOCAL BEST Group Iași	Iași, RO
IT Department Coordinator	Sep 2023 – Aug 2024
- Directed IT operations and maintained technical infrastructure across organizational events. - Mentored volunteers and oversaw delivery of technical projects. - Administered hardware, software licensing, and network resources.	
Event Coordinator — JobShop Career Event	Nov 2022 – Apr 2023
- Coordinated logistics and operations for a large student–industry career event. - Managed scheduling, partner engagement, and on-site execution. - Improved attendee experience through strategic planning and workflow optimization.	
TOPICS Responsible — Prove It 24H Techathon	Dec 2021 – Mar 2022
- Defined technical requirements and coordinated hackathon logistics. - Supported participants with rules, resources, and real-time assistance. - Facilitated communication between organizers, industry mentors, and teams.	
IT Responsible — BEST Training Weekend	Jan 2021 – May 2022
- Developed and maintained the event website using HTML, CSS, and JavaScript. - Collaborated with design and marketing responsables on content, branding, and technical setup.	

ADDITIONAL SKILLS

AI / ML:	Machine Learning, Deep Learning, LLMs, RAG Systems, PyTorch, Prompt Engineering
Web Apps:	React, TypeScript, Node.js, Firebase, PostgreSQL, WordPress
Soft Skills:	Problem Decomposition, Clear Communication, Knowledge Sharing, Self-directed Learning